

Memo

To: EDRM 718 Analysts

From: Mike Seaman

Date: 24 April 2025

Re: Two Useful Functions

As you know, we have been building our company library of useful functions on an as-needed basis when we encounter projects that can benefit from a new function. The recent contract from RWS has led me to consider two additional functions that should prove useful for this project, as well as future projects. I have described these below. Please incorporate these into your RWS project repository. Alternatively, you may choose to use the functions that I have already written, though this will reduce your level of compensation, as stated in your annual employment agreement. If you choose to use my functions, please email me a request.

1. Researchers in many disciplines use surveys based on the Likert scale. A survey item will ask the respondent to respond on a specified scale (such as 1 to 6) to indicate where the respondent self-reports level of intensity when measuring an attribute such as agreement, preference, or importance. The survey consists of multiple items and the total score is based on summing across items. This score is meant to reflect the respondent's placement on a construct scale. Two examples would be a survey to assess beliefs about gun control or a survey to assess desire to obtain an advanced educational degree. To increase the validity of the scale, some items should be written so that the scale is reversed. That is, a high score on the item will make a small contribution to the total score and a low score will make a large contribution. Such items are called "reverse scored" items. For example, on a six-point scale, a response of "2" will be scored as "5."

Write a function that will compute the total score for respondents on a Likert scale. Your function should accept three required arguments: a data frame with the individual item responses (as usual, individuals in rows and items in columns), a numeric with the scale used for the items (e.g., you would send "6" if the items were on a six-point scale), and a numeric vector indicating which item numbers should be reverse scored. Your function should return a data frame that is the original data frame with an additional column added to the end that contains the score for each respondent.

2. Write a function that accepts (1) a quantitative response variable, (2) a quantitative explanatory variable, and (3) a number. Suppose we denote the response variable as y , the explanatory variable as x , and the number as p . Your function should model y as a linear function of x and then for each (x,y) pair determine if the residual is one of the p largest in the set of residuals, with largest defined in terms of absolute value (e.g., a residual of -10 is just as large as a residual of 10). The function should return a logical vector indicating TRUE for each pair that has one of the p th largest residuals and FALSE otherwise.